

ECON3140_empirical project hw

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```
knitr::opts_chunk$set(message = FALSE, warning = FALSE)
```

```
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.1      v stringr    1.5.1
```

```
## v ggplot2    3.5.2      v tibble     3.2.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.1
```

```
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(sandwich)
```

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'zoo'
```

```
##
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
library(stargazer)
```

```
##
```

```
## Please cite as:
```

```
##
```

```
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
```

```
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
library(readxl)
```

Question 1

The paper studies whether racial discrimination exists in mortgage lending. The authors specifically look at whether Black and Hispanic mortgage applicants are denied loans more often than similarly qualified white applicants. The data comes from the 1990 Home Mortgage Disclosure Act (HMDA) reports for the Boston metropolitan area. The original HMDA data included details such as applicant race, income, loan amount, and whether the loan was denied or approved. However, critics said the data were incomplete because they left out important lending factors like credit history, debt levels, loan-to-value ratios, employment history, and liquid assets. Earlier studies found that minorities were denied mortgages at much higher rates than whites. Lenders argued that these differences were due to missing financial characteristics rather than discrimination. To tackle this criticism, the Federal Reserve Bank of Boston collected many additional underwriting factors directly from lenders. The authors then estimated mortgage denial models while considering these extra risk factors. The results showed that minorities generally had weaker financial profiles than whites. These factors explained some of the racial gap in denial rates. However, race still remained an important factor even after controlling for many additional financial and credit variables. The paper concludes that racial discrimination likely played a significant role in mortgage lending decisions.

```
hmda <- read_excel("~/Desktop/academics/spring 2026/ECON 3140/hw/data/HMDA-excel.xls")

hmda[hmda == 999999.4] <- NA

# denied dummy
hmda$denied <- 0
hmda$denied[hmda$s7 == 3] <- 1

# black dummy
hmda$black <- 0
hmda$black[hmda$s13 == 3] <- 1

# payment-income ratio
hmda$PI <- hmda$s46

# housing-income ratio
hmda$HI <- hmda$s45

# loan-to-value ratio
hmda$LV <- hmda$s6 / hmda$s50

# square term
hmda$LV2 <- hmda$LV^2

# credit score variables
hmda$CCS <- hmda$s43
hmda$MCS <- hmda$s42

# PMI denied
hmda$NoMI <- hmda$s53

# self-employed
hmda$self <- hmda$s27a
```

Question 2

The assignment first requires creating indicator variables for mortgage denial and applicant race. The variables:

```
# hmda$denied <- 0
# hmda$denied[hmda$s7==3] <- 1
```

creates a dummy variable equal to 1 if the mortgage applicant was denied and 0 otherwise. The variable:

```
# hmda$black <- 0
# hmda$black[hmda$s13==3] <- 1
```

creates a dummy variable equal to 1 for Black applicants because the codebook defines s13=3 as Black. The remaining variables correspond to financial characteristics used in mortgage underwriting, including debt burden, loan-to-value ratios, credit history, and employment status.

Question 3

```
# cross-tabulation
table(hmda$black, hmda$denied)
```

```
##
##      0      1
## 0 1852  189
## 1  243   96
```

```
# rejection rates
prop.table(table(hmda$black, hmda$denied), 1)
```

```
##
##      0      1
## 0 0.90739833 0.09260167
## 1 0.71681416 0.28318584
```

```
# group means
white.rate <- mean(hmda$denied[hmda$black==0], na.rm=TRUE)
black.rate <- mean(hmda$denied[hmda$black==1], na.rm=TRUE)
white.rate
```

```
## [1] 0.09260167
```

```
black.rate
```

```
## [1] 0.2831858
```

```
denied = B0 + B1 black + u
B0.hat = mean denial rate for whites
B1.hat = difference in denial rates
```

```
beta0.hat <- white.rate
beta1.hat <- black.rate - white.rate
beta0.hat
```

```
## [1] 0.09260167
```

```
beta1.hat
```

```
## [1] 0.1905842
```

```
reg3 <- lm(denied ~ black, data=hmda)
summary(reg3)
```

```
##
## Call:
## lm(formula = denied ~ black, data = hmnda)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.2832 -0.0926 -0.0926 -0.0926  0.9074
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.092602   0.007037   13.16  <2e-16 ***
## black        0.190584   0.018644   10.22  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3179 on 2378 degrees of freedom
## Multiple R-squared:  0.04209,    Adjusted R-squared:  0.04169
## F-statistic: 104.5 on 1 and 2378 DF,  p-value: < 2.2e-16
```

$\text{denied} = 0.09260167 + 0.1905842 \text{ black} + u$

The estimated denial rate for white applicants is approximately 9.3%, while the estimated denial rate for Black applicants is approximately 28.3%. The estimated coefficient on black is approximately 0.191. This means Black applicants have about a 19.1 percentage-point higher probability of mortgage denial than white applicants. The coefficient is highly statistically significant with a p-value less than 0.001. Therefore, there is strong evidence of a difference in denial rates across racial groups in the sample.

Question 4

```
reg4 <- lm(denied~PI, data=hmnda)
summary(reg4)
```

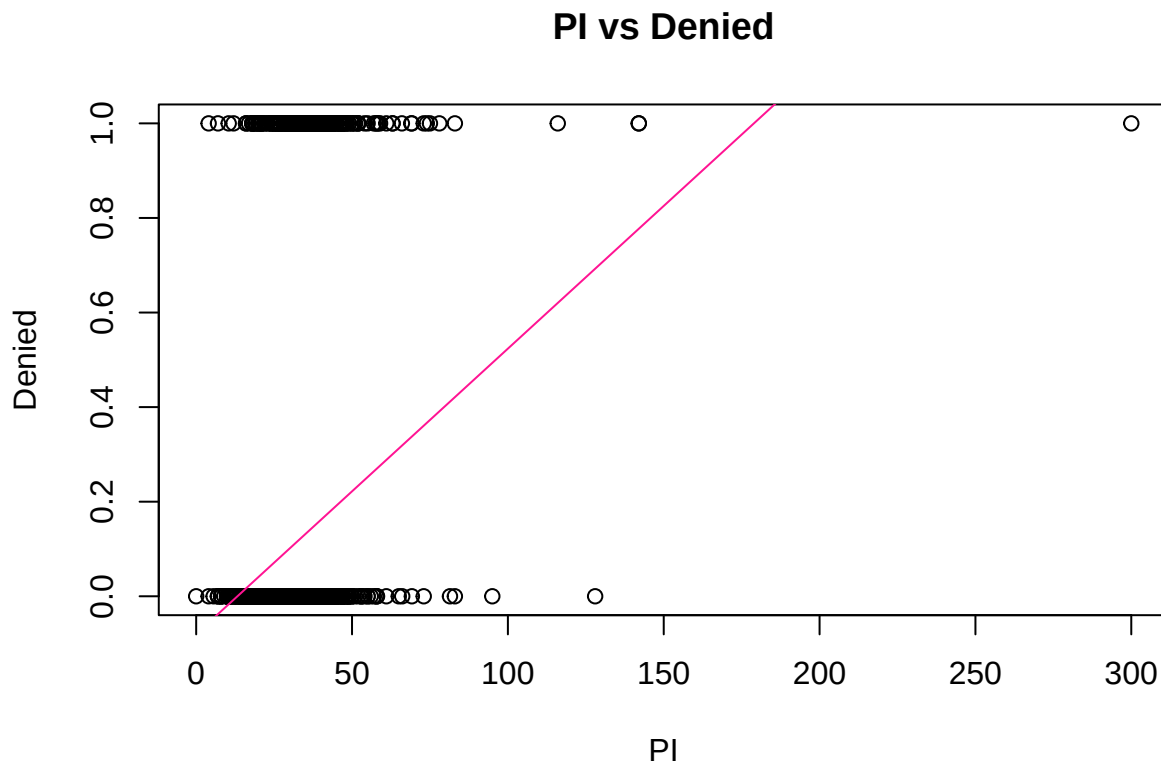
```
##
## Call:
## lm(formula = denied ~ PI, data = hmnda)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.73070 -0.13736 -0.11322 -0.07097  1.05577
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0799096  0.0211578  -3.777 0.000163 ***
## PI           0.0060353  0.0006084   9.920  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3183 on 2378 degrees of freedom
## Multiple R-squared:  0.03974,    Adjusted R-squared:  0.03933
## F-statistic: 98.41 on 1 and 2378 DF,  p-value: < 2.2e-16
```

```
# robust SE
coeftest(reg4, vcov=vcovHC(reg4, type="HC1"))
```

```
##
## t test of coefficients:
```

```
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.07990965  0.03196661 -2.4998  0.01249 *
## PI          0.00603535  0.00098483  6.1283 1.036e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# scatterplot
plot(hmda$PI, hmda$denied, xlab="PI", ylab="Denied", main="PI vs Denied")
abline(reg4, col="deeppink")
```



$$\text{denied} = -0.0799096 + 0.0060353 \text{ PI}$$

The estimated coefficient on PI is approximately 0.0060 and is highly statistically significant. This means that a one-unit increase in the payment-to-income ratio increases the predicted probability of mortgage denial by about 0.6 percentage points. The positive relationship indicates that applicants with larger payment burdens are more likely to be denied mortgages. Using robust standard errors is appropriate because the dependent variable is binary, which creates heteroskedasticity in the error term of the linear probability model. The fitted regression line in the scatterplot slopes upward, which is consistent with the positive estimated coefficient and the economic intuition that higher debt burdens increase lending risk.

Question 5

```
predict(reg4, newdata=data.frame(PI=20))
```

```
##          1  
## 0.04079734
```

```
predict(reg4, newdata=data.frame(PI=10))
```

```
##          1  
## -0.01955615
```

The predicted values represent estimated probabilities of mortgage denial. For $PI = 20$, the model predicts approximately a 4.1% probability of denial. For $PI = 10$, the model predicts a negative probability, which is impossible in reality because probabilities must lie between 0 and 1. This reveals one weakness of the linear probability model. Because OLS is linear, fitted values can fall outside the valid probability range. This motivates using nonlinear probability models such as logit and probit.

Question 6

```
# logit model  
logit6 <- glm(denied~PI, data=hmda, family=binomial(link="logit"))  
summary(logit6)
```

```
##  
## Call:  
## glm(formula = denied ~ PI, family = binomial(link = "logit"),  
##      data = hmda)  
##  
## Coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept) -4.028432   0.268576 -14.999 < 2e-16 ***  
## PI           0.058845   0.007336   8.021 1.05e-15 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for binomial family taken to be 1)  
##  
##    Null deviance: 1744.2  on 2379  degrees of freedom  
## Residual deviance: 1660.2  on 2378  degrees of freedom  
## AIC: 1664.2  
##  
## Number of Fisher Scoring iterations: 5
```

```
# predicted probabilities  
predict(logit6, newdata=data.frame(PI=20), type="response")
```

```
##          1  
## 0.05460215
```

```
predict(logit6, newdata=data.frame(PI=10), type="response")
```

```
##          1  
## 0.03106901
```

In the logit model, the estimated coefficient on PI is approximately 0.0588 and is highly statistically significant. The positive coefficient indicates that larger payment-to-income ratios increase the likelihood of mortgage

denial. Unlike the linear probability model, the logit model constrains fitted probabilities to remain between 0 and 1. The predicted probability of denial is approximately 5.5% when PI = 20 and approximately 3.1% when PI = 10. These fitted values are economically meaningful because they remain within the valid probability range.

Question 7

```
# OLS regression
ols7 <- lm(denied~PI + black, data=hmda)
summary(ols7)

##
## Call:
## lm(formula = denied ~ PI + black, data = hmda)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.62526 -0.11772 -0.09293 -0.05488  1.06815
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0905136  0.0207878  -4.354 1.39e-05 ***
## PI           0.0055919  0.0005987   9.340 < 2e-16 ***
## black        0.1774282  0.0183694   9.659 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3123 on 2377 degrees of freedom
## Multiple R-squared:  0.076, Adjusted R-squared:  0.07523
## F-statistic: 97.76 on 2 and 2377 DF, p-value: < 2.2e-16

coeftest(ols7, vcov=vcovHC(ols7, type="HC1"))

##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -0.09051360  0.02859959 -3.1649  0.001571 **
## PI           0.00559195  0.00088666  6.3067 3.387e-10 ***
## black        0.17742824  0.02494627  7.1124 1.502e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# logit regression
logit7 <- glm(denied~PI+black, data=hmda, family=binomial(link="logit"))
summary(logit7)

##
## Call:
## glm(formula = denied ~ PI + black, family = binomial(link = "logit"),
##      data = hmda)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.125558  0.268414 -15.370 < 2e-16 ***
```

```
## PI          0.053704    0.007283    7.374 1.66e-13 ***
## black       1.272782    0.146198    8.706 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1744.2  on 2379  degrees of freedom
## Residual deviance: 1591.4  on 2377  degrees of freedom
## AIC: 1597.4
##
## Number of Fisher Scoring iterations: 5

OLS: denied = -0.0905136 + 0.00559195 PI + 0.17742824 black + u
logit: denied = -4.125558 + 0.053704 PI + 1.272782 black + u
```

After controlling for payment burden, the coefficient on black remains positive and highly statistically significant in both the OLS and logit models. In the OLS model, the estimated coefficient on black is approximately 0.177. This means Black applicants are predicted to have about a 17.7 percentage-point higher probability of denial than otherwise similar white applicants with the same payment-to-income ratio. The estimated coefficient on PI is approximately 0.0056, which indicates that higher payment burdens also increase denial probabilities. In the logit model, the coefficient on black is approximately 1.273 and remains highly statistically significant. Although logit coefficients are not interpreted directly as probability changes, the positive coefficient still indicates that Black applicants face substantially higher predicted denial probabilities. These results suggest that differences in payment burden alone cannot fully explain the racial gap in mortgage denials.

Question 8

The coefficient on black should not automatically be interpreted as the causal effect of racial discrimination. There may still be omitted variables that are correlated with both race and mortgage denial. Examples include: credit history, wealth, employment stability, loan-to-value ratios, neighborhood risk, and liquid assets. If these variables are omitted from the regression, the coefficient on black may partly capture differences in financial characteristics rather than discrimination itself. This omitted-variable problem was one of the major criticisms of earlier HMDA studies discussed by Munnell et al. The original HMDA data lacked many important underwriting variables, making it difficult to separate discrimination from differences in borrower risk.

Question 9

```
# OLS
ols9 <- lm(denied~ black + PI + HI + LV + LV2 + CCS + MCS + NoMI + self, data=hmda)
summary(ols9)

##
## Call:
## lm(formula = denied ~ black + PI + HI + LV + LV2 + CCS + MCS +
##      NoMI + self, data = hmda)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.14414 -0.12296 -0.05789 -0.00753  1.09839
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -0.2185699 0.0485854 -4.499 7.17e-06 ***
## black      0.1024729 0.0174053  5.887 4.48e-09 ***
## PI         0.0050429 0.0008852  5.697 1.37e-08 ***
## HI        -0.0006241 0.0009795 -0.637 0.524089
## LV        -0.0499138 0.1221493 -0.409 0.682849
## LV2       0.1130415 0.0859053  1.316 0.188340
## CCS       0.0391006 0.0036461 10.724 < 2e-16 ***
## MCS       0.0229060 0.0112694  2.033 0.042206 *
## NoMI      0.7275353 0.0421847 17.246 < 2e-16 ***
## self      0.0658970 0.0183216  3.597 0.000329 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2846 on 2370 degrees of freedom
## Multiple R-squared:  0.235, Adjusted R-squared:  0.2321
## F-statistic: 80.88 on 9 and 2370 DF, p-value: < 2.2e-16
```

```
coeftest(ols9, vcov=vcovHC(ols9, type="HC1"))
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.21856993 0.07455102 -2.9318 0.003402 **
## black      0.10247293 0.02294879  4.4653 8.371e-06 ***
## PI         0.00504290 0.00116806  4.3173 1.644e-05 ***
## HI        -0.00062406 0.00115258 -0.5414 0.588252
## LV        -0.04991377 0.22682179 -0.2201 0.825846
## LV2       0.11304146 0.17966878  0.6292 0.529301
## CCS       0.03910056 0.00476360  8.2082 3.643e-16 ***
## MCS       0.02290596 0.01158434  1.9773 0.048121 *
## NoMI      0.72753531 0.04177732 17.4146 < 2.2e-16 ***
## self      0.06589703 0.02097296  3.1420 0.001699 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# logit
```

```
logit9 <- glm(denied ~ black + PI + HI + LV + LV2 + CCS + MCS + NoMI + self, data=hmda, family=binomial)
summary(logit9)
```

```
##
## Call:
## glm(formula = denied ~ black + PI + HI + LV + LV2 + CCS + MCS +
##      NoMI + self, family = binomial(link = "logit"), data = hmda)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -7.284695  0.877711 -8.300 < 2e-16 ***
## black      0.800842  0.169847  4.715 2.42e-06 ***
## PI         0.049903  0.010360  4.817 1.46e-06 ***
## HI        -0.002647  0.012138 -0.218 0.827367
## LV         2.582210  1.923354  1.343 0.179416
## LV2       -0.429369  1.170556 -0.367 0.713763
## CCS       0.346249  0.037558  9.219 < 2e-16 ***
## MCS       0.317344  0.138393  2.293 0.021845 *
```

```
## NoMI          4.505393    0.549269    8.203 2.35e-16 ***
## self          0.719261    0.206986    3.475 0.000511 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1744.2  on 2379  degrees of freedom
## Residual deviance: 1315.6  on 2370  degrees of freedom
## AIC: 1335.6
##
## Number of Fisher Scoring iterations: 6

OLS: denied = -0.2185699 + 0.1024729 black + 0.0050429 PI - 0.0006241 HI - 0.0499138 LV + 0.1130415
LV2 + 0.0391006 CCS + 0.0229060 MCS + 0.7275353 NoMI + 0.0658970 self

logit: denied = -7.284695 + 0.800842 black + 0.049903 PI - 0.002647 HI + 2.582210 LV - 0.429369 LV2 +
0.346249 CCS + 0.317344 MCS + 4.505393 NoMI + 0.719261 self
```

In the full OLS specification, the estimated coefficient on black is approximately 0.102 and remains highly statistically significant. This means that, even after controlling for debt burden, credit history, loan-to-value ratios, PMI denial, and self-employment status, Black applicants are associated with about a 10.2 percentage-point higher probability of mortgage denial in the linear probability model, holding other variables constant. The magnitude of the race coefficient declines relative to simpler specifications, suggesting that observable financial differences explain part of the racial gap in denial rates. However, the coefficient remains positive and statistically significant, indicating that the gap is not fully explained by controls.

Several financial variables are also statistically significant:

1. PI has a positive coefficient of approximately 0.0050. A higher payment-to-income ratio is associated with a higher probability of denial, consistent with the idea that higher debt burden increases default risk.
2. CCS has a positive coefficient of approximately 0.039.
3. MCS has a positive coefficient of approximately 0.023.

Worse credit scores (higher values) are strongly associated with higher denial probabilities, as expected.

4. NoMI has a very large positive coefficient of approximately 0.728. The large positive coefficient on NoMI indicates that applicants who were denied mortgage insurance are substantially more likely to be denied a mortgage.

5. self has a positive coefficient of approximately 0.066. Self-employment is associated with a modest increase in denial probability, consistent with potentially higher income uncertainty.

HI, LV, and LV² are not statistically significant in this specification, suggesting that once credit history and other risk measures are included, leverage and housing expense ratios do not have independent predictive power for mortgage denial. In the logit model, the coefficient on black is approximately 0.801 and remains statistically significant. Overall, the results are consistent with the findings of Munnell et al.: financial characteristics explain part of the racial gap in mortgage denials, but race still remains statistically significant after extensive controls are added.

Question 10

```
# race effect in OLS
coef(ols9)["black"]
```

```
##      black
## 0.1024729
```

The coefficient on black in the OLS model is approximately 0.102. This means that, holding all other explanatory variables constant, Black applicants are predicted to have about a 10.2 percentage-point higher probability of mortgage denial than otherwise similar white applicants. The coefficient remains statistically significant even after controlling for debt burden, credit history, PMI denial, and employment characteristics.

Question 11

```
library(margins)
summary(margins(logit9))
```

##	factor	AME	SE	z	p	lower	upper
##	black	0.0630	0.0133	4.7502	0.0000	0.0370	0.0890
##	CCS	0.0272	0.0029	9.3120	0.0000	0.0215	0.0330
##	HI	-0.0002	0.0010	-0.2181	0.8274	-0.0021	0.0017
##	LV	0.2031	0.1514	1.3409	0.1799	-0.0937	0.4999
##	LV2	-0.0338	0.0921	-0.3667	0.7138	-0.2142	0.1467
##	MCS	0.0250	0.0109	2.2927	0.0219	0.0036	0.0463
##	NoMI	0.3543	0.0417	8.5045	0.0000	0.2726	0.4359
##	PI	0.0039	0.0008	4.8419	0.0000	0.0023	0.0055
##	self	0.0566	0.0163	3.4756	0.0005	0.0247	0.0885

In a logit model, coefficients represent changes in the log-odds of denial and therefore cannot be interpreted directly as changes in probability. To obtain economically meaningful effects, we compute marginal effects, which translate changes in explanatory variables into changes in predicted probabilities. The average marginal effect of black is approximately 0.063, meaning that being Black increases the predicted probability of denial by about 6.3 percentage points on average, holding other variables constant. The marginal effect is smaller than the OLS estimate because the logit model is nonlinear and constrains predicted probabilities to remain between 0 and 1.

Question 12

```
# predicted effects for average applicant
```

```
avg.applicant <- data.frame(black=0,
                             PI=mean(hmda$PI, na.rm=TRUE),
                             HI=mean(hmda$HI, na.rm=TRUE),
                             LV=mean(hmda$LV, na.rm=TRUE),
                             LV2=mean(hmda$LV2, na.rm=TRUE),
                             CCS=mean(hmda$CCS, na.rm=TRUE),
                             MCS=mean(hmda$MCS, na.rm=TRUE),
                             NoMI=mean(hmda$NoMI, na.rm=TRUE),
                             self=mean(hmda$self, na.rm=TRUE))
```

```
# white applicant
```

```
avg.applicant$black <- 0
pred.white <- predict(logit9, newdata=avg.applicant, type="response")
```

```
# black applicant
```

```
avg.applicant$black <- 1
pred.black <- predict(logit9, newdata=avg.applicant, type="response")
```

```
pred.white
```

```
##          1
## 0.06977031
pred.black
```

```
##          1
## 0.1431486
```

```
pred.black - pred.white
```

```
##          1  
## 0.0733783
```

This calculation compares predicted denial probabilities for two otherwise identical applicants who differ only in race. The predicted probability of denial is approximately 7.0% for the average white applicant and 14.3% for the average Black applicant. The difference implies that, for an otherwise identical applicant, being Black is associated with an approximately 7.3 percentage-point higher probability of denial. The difference between the OLS estimate and the logit marginal effect arises because the logit model is nonlinear and estimates effects on probabilities rather than a linear approximation.

Overall Conclusion

The empirical results show that: financial characteristics strongly affect mortgage approval decisions, minority applicants generally have weaker financial profiles, controlling for these factors substantially reduces the racial gap, but race still remains statistically significant. This supports the conclusion of Munnell et al. that racial disparities in mortgage lending cannot be fully explained by observable financial characteristics alone.